

MTE544 – Autonomous Mobile Robotics

University of Waterloo

Department of Mechanical and Mechatronics Engineering

Lab 1: Sensor data processing for mobile robots

Group member (Family Name, First Name)	**Student ID**
Adams, Hunter	20997503
Chen, Joe	20998545
Wielowieyski, Jacob	21004619

Station number: _____

Robot number: 7

October 19, 2026

1 Odometry results and discussion

Figure 1 retains all logged poses. For each file, $t = (s - s_0)10^{-9}$ s converts nanosecond stamps to elapsed time. Positions remain in the odometry frame; yaw is unwrapped to remove 2π representation jumps.

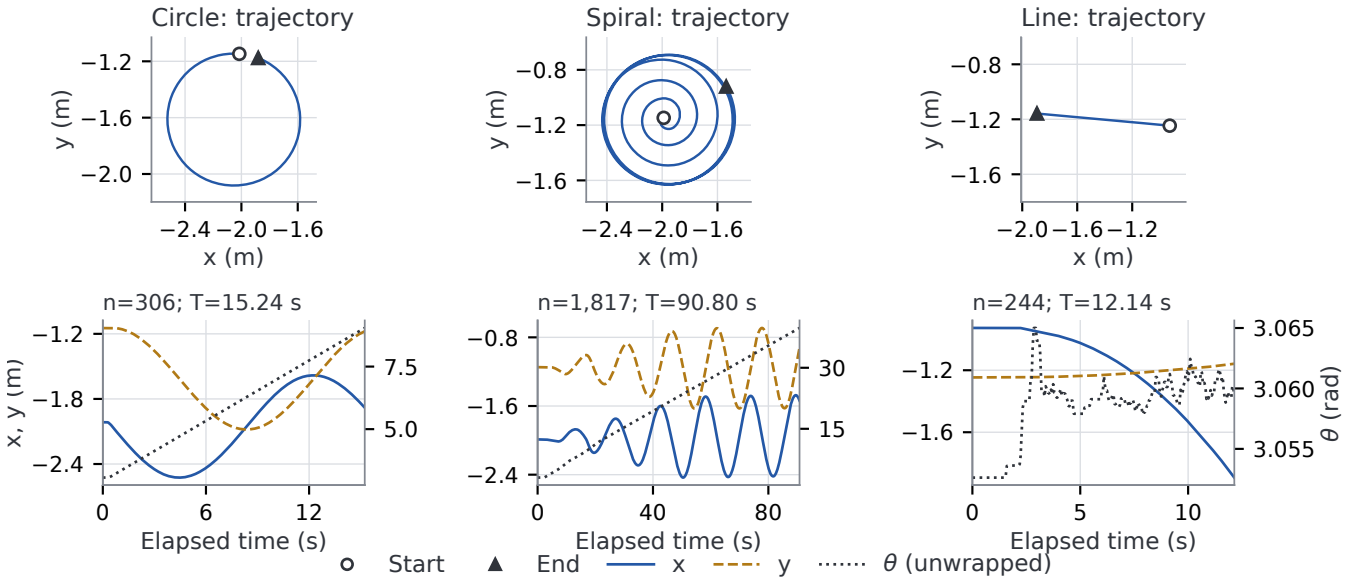


Figure 1: Odometry: equal-aspect x–y paths (top) and position/heading histories (bottom). Left axes show metres; right axes show unwrapped radians. Each run uses its own axis ranges. Titles give sample count n and duration T.

Circle: The resulting radius is 0.470 m; θ increased by 5.98 radians during the 15.24 s recording. Since this is less than 2π radians, a full circle was not completed.

Spiral: The outer-loop radius is approximately 0.470 m; θ increased by 36.79 radians during the 90.80 s recording. The loops initially widen, then overlap near the outer boundary as the radius becomes approximately constant.

Line: The path length is 0.970 m over 12.14 s, with heading variation of only 0.0124 rad (0.71°). The increasing magnitude of the position slope indicates acceleration.

2 IMU results and discussion

Figure 2 shows logged IMU-frame accelerations (specific force) and z-axis angular rate.

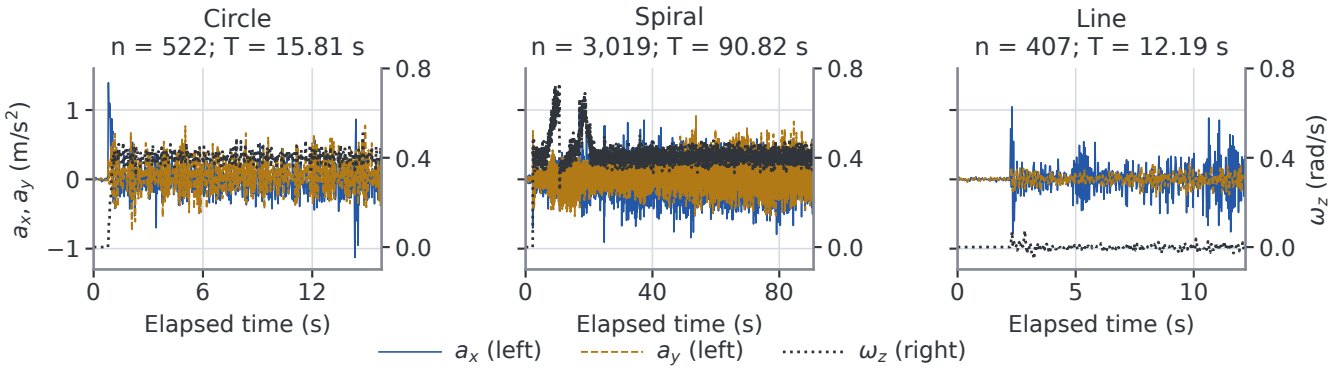


Figure 2: Unsmoothed IMU samples on shared scales. Acceleration uses the left axes and angular rate the right; elapsed time starts at each file's first timestamp.

Circle: Mean ω_z is 0.402 rad/s over 2–15 s of the 15.81 s recording, showing sustained turning after startup.

Spiral: Mean ω_z is 0.403 rad/s over 25–90 s of the 90.82 s recording; larger early variations settle into steady turning.

Line: Mean ω_z is 5.4×10^{-4} rad/s over 12.19 s, indicating little net rotation. The mean of a_x is close to zero (0.0017 m/s^2), indicating that positive and negative readings largely balance over the recording.

3 LiDAR results and map acquisition

Figure 3 uses the scan nearest each run's temporal midpoint. Rows contain 720 ranges followed by angle metadata and timestamp. Using original beam indices, $\alpha_i = \alpha_{\min} + i\Delta\alpha$, $x_i = r_i \cos\alpha_i$, and $y_i = r_i \sin\alpha_i$. Removing positive-infinity returns retains 456, 458, and 488 points. All finite positive ranges remain because sensor limits were not logged. Wall-like segments appear in every scan; their orientations differ in these local sensor frames.

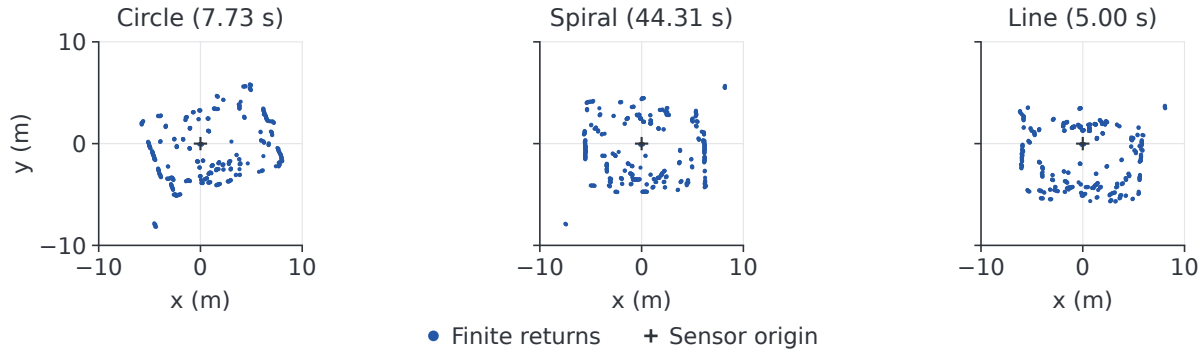


Figure 3: Midpoint LiDAR scans in their individual sensor frames; + marks the sensor origin. Times are relative to each LiDAR log's first timestamp.

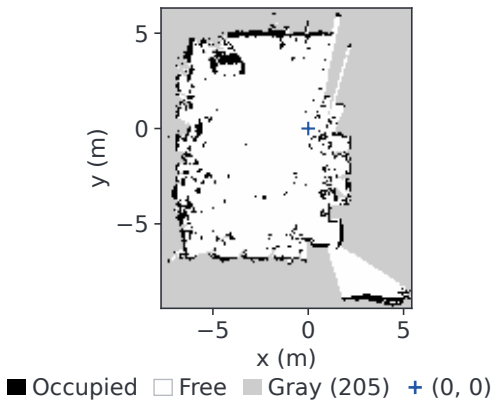


Figure 4 preserves the saved map's pixels. A broad free region and room boundary are visible, but scattered returns and thick, broken boundaries limit geometric precision. Grey unknown/intermediate cells occupy 49.3% of the image. The YAML free threshold, 0.25, exceeds grey-205 occupancy $50/255 \approx 0.196$, so these cells would reload as free; this metadata limitation requires correction before reuse.

Figure 4: Saved PGM map, 263×315 cells at 0.05 m/cell.
Lower-left origin: $(-7.72, -9.43)$ m, zero yaw.

Overall Interpretation:

- **Odometry:** Outputs position and heading, showing a partial circle, widening spiral loops that later overlap, and a nearly straight path with increasing speed.
- **IMU:** Outputs sensor-frame acceleration and angular rate, indicating turning during the circle and spiral, and little net turning during the line.
- **LiDAR:** Outputs distances at known beam angles, showing nearby surfaces. The saved map shows free space and obstacles in the environment.